Math 340 • Fall 2026 • Victor Matveev

Lab Project #8

Numerical Differentiation (section 5.1)

Finite difference approximations generally become more accurate as the step size $\boldsymbol{h}$ decreases, until roundoff errors begin to affect the results. In this lab, you will examine how the approximation error depends on $\boldsymbol{h}$.

We will use the centered finite difference approximation of the first derivative

$$\boldsymbol{D(h) \equiv \frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{f'''(c)}{6}h^2}, \text{ where } c \in (x-h, x+h)$$

For small *h*, the theoretical absolute truncation error is therefore approximately equal $\boldsymbol{E_T(h) \approx \frac{1}{6}|f'''(x)|h^2}$.

Part 1. MATLAB function

Write a function that computes the above centered finite difference $\boldsymbol{D(h)}$ for $\boldsymbol{h = 10^{-p}}$, where $\boldsymbol{p = 1, 2, .. 15}$.

Your MATLAB function should accept the following arguments:

- "Handle" of the functions $f(x)$, $f'(x)$, $f'''(x)$.
- The value of x at which the derivative is evaluated.

The MATLAB function should do the following:

1. For each value $h = 10^{-p}$ (where $p = 1 \ldots 15$), print:
    - The centered finite difference approximation, $D(h)$
    - Its actual absolute error $|D(h) - f'(x)|$
    - The approximate theoretical truncation error, $\frac{1}{6}|f'''(x)|h^2$.
2. Plot the following three quantities against *h* on the same log-log graph, with a legend:
    - The actual absolute error
    - The theoretical truncation error
    - The quantity $\boldsymbol{\epsilon_M|f(x)|/h}$, where $\epsilon_M = 2^{-52}$ is known as the "machine epsilon".

    Hint: look up loglog command for log-scale plots.
3. Print the value of $\boldsymbol{h}$ that minimizes the actual error among the tested values.

Part 2. Numerical experiments

Run your code with the following functions:

1. $f(x) = e^x$, at $x = 0$.

2. $f(x) = \sin x$, at $x = 1$.

For each case, use your error plot to answer the following questions:

1. Compare the optimal values of $\boldsymbol{h}$ for the two functions.
2. Why does the actual error initially decrease as $\boldsymbol{h}$ becomes smaller, but eventually increase?
3. Does your plot approximately support the prediction that the roundoff error is proportional to $\boldsymbol{\epsilon_M|f(x)|/h}$?